



Federal Urdu University
of Arts, Science and Technology

ARTIFICIAL INTELLIGENCE

ASSIGNMENT - 1

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AI FOOD RECOMMENDATION AGENT

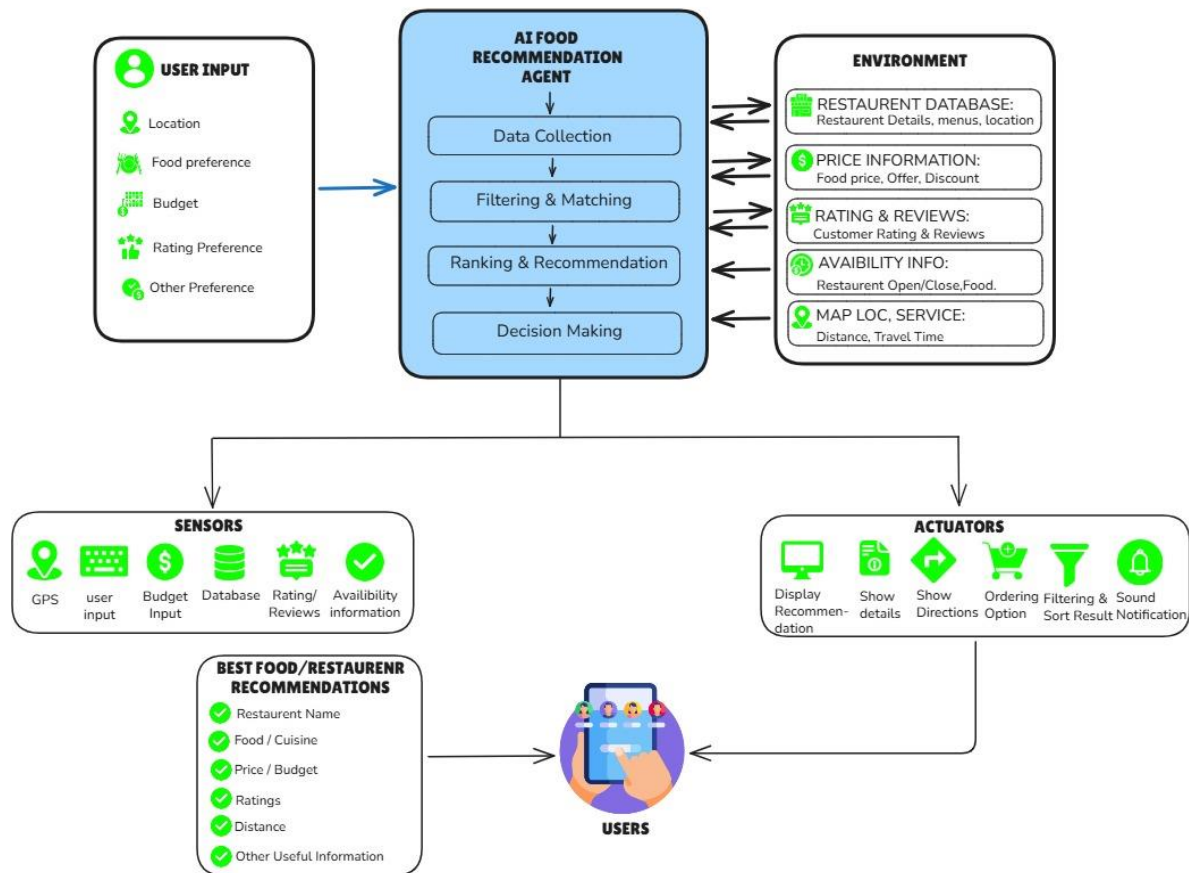
Introduction:

My agent helps users find food or restaurants according to their location, budget and food choice.

For example, a user can say: "I want Pakistani food near me under Rs. 1500."

The agent will suggest suitable restaurants.

Graphical Representation:



Environment

The environment includes restaurants, food, prices, locations, ratings and availability.

Environment Type:

Partially Observable : It does not know everything about the user.

Non-Deterministic: Prices and availability can change.

Dynamic: Restaurant information can change with time.

Mostly Discrete: It selects from available food/restaurants.

Multi-Agent: It can interact with users, restaurants and map services.

Sensors

The agent gets information through:

GPS/location

User input

Budget

Food preferences

Restaurant data

Ratings and reviews

Actuators

The agent can:

Show restaurants

Show food and prices

Show ratings and distance

Give directions

Provide ordering options

Deterministic / Non-Deterministic

It is Non-Deterministic because restaurant prices, availability and ratings can change.

Static / Dynamic

It is Dynamic because the user's location and restaurant information can change at any time.

Single / Multi-Agent

It is a Multi-Agent Environment because it works with users, restaurants, GPS/maps and food delivery services.

Learning

The agent can learn from the user's previous choices. For example, if a user mostly chooses Pakistani food under Rs. 1500, it can suggest similar options next time.

PEAS

P – Performance: Good and suitable food recommendations.

E – Environment: Restaurants, food, prices, users and locations.

A – Actuators: Recommendations, directions and ordering options.

S – Sensors: GPS, user input, restaurant data, ratings and reviews.